Understanding Bilinear Value Functions in Goal-Conditioned RL: a Socratic Record of Every Experiment

Living document — v1, 2026-07-16

Factored-FB project

Method of this document: each paragraph poses a question, describes the experiment run to answer it (with training details at conference-paper level), and states the findings. The chain continues until the only remaining questions are the currently unanswerable ones about why bilinear Q-networks work — collected in §3 with proposed experiments. §4 surveys bilinearity and factored value representations in RL. Companion: `fb-variants-and-grounding.tex`.

1 Common experimental substrate

Benchmark. OGBench `cube-single-play-v0`: a 6-DoF arm, one cube, 28-D state observations, 5-D actions in $[-1, 1]$; offline dataset of 1M transitions from a scripted play policy (repeated pick-and-place at random locations). Evaluation: the 5 `singletask` goals, horizon 200, success = cube within 4 cm of the target (the environment's criterion). Unless stated: mean success over 50 episodes/task, 3–13 seeds, 500k gradient steps. Cross-domain cells use `antmaze-medium-navigate-v0` and `scene-play-v0`.

Value learner (GC-IQL). Rewards $r \in \{-1, 0\}$ (0 at the goal), masks $m = 1 - \mathbf{1}[s'{=}g]$; expectile value loss and MSE critic loss

$$\mathcal{L}_V = \mathbb{E}\,\ell_\tau^2\big(\min_i \bar{Q}_i(s,a,g) - V(s,g)\big), \quad \mathcal{L}_Q = \sum_i \mathbb{E}\big(Q_i(s,a,g) - (r + \gamma m V(s',g))\big)^2, \tag{1}$$

τ=0.9, γ=0.99, twin critics, target EMA τ_{ema}=0.005 (critic only). Networks: 3-layer 512 MLPs, GELU + LayerNorm; Adam $3{\times}10^{-4}$, batch 256. Goal relabeling (value loss): 20% current / 50% geometric future (p=1−γ) / 30% random; actor goals: uniform future on the same trajectory.

The two heads under comparison.

$$\text{monolithic (MLP): } Q_\theta([s,a,g]) \qquad \text{vs.} \qquad \text{bilinear: } Q(s,a,g) = \frac{\phi(s,a)^\top \psi(g)}{\sqrt{d}},\ d{=}50. \tag{2}$$

Both use the same trunk widths; the bilinear head has $\approx 2\times$ the trunk parameters (two towers) — a confound with a queued control (§3).

Policy extraction. (i) *Flow* (FQL-style): a goal-agnostic BC flow $v(s, x_t, t)$ trained by flow matching; a one-shot actor $\pi(s, c, \varepsilon)$ trained by $-\min_i Q_i/\text{sg}|Q| + 3.0\,\|\pi - \text{Euler}_{10}(v)\|^2$. (ii) *DDPG+BC*: Gaussian policy, $-Q/\text{sg}|Q| - \alpha \log \pi_\beta$, α=0.3.

2 The Socratic chain

Q: Why do unsupervised/zero-shot RL methods fail on manipulation at all?
Experiment. Five methods — FB, PSM, RLDP, CRL, and GCIQL as the supervised goal-conditioned

anchor — each trained on the same dataset (10 seeds, 1M steps for the torch cells), plus a *phase diagnostic* that segments every evaluation rollout into approach → control acquisition → maintained control → transport using cube–effector geometry and the contact channel (3 seeds × 100 episodes per method). *Findings.* GCIQL 0.72 ≫ FB 0.42–0.52 > PSM 0.36 > CRL 0.31 > RLDP 0.17. Approach failures are negligible for every method (P(reach)=1.0); grasping differs by ≤ 6 points; **the entire method spread is post-grasp**: P(success | secured) = FB 0.44 / MLP-GCIQL 0.80 / bilinear-GCIQL 0.98. The object to explain is per-step action selection while in control, not reaching or grasping.

Q: Why does FB fail — architecture or training objective?

Experiment. The *graft lattice*: hold the substrate of §1 fixed and swap single ingredients between FB and GCIQL (3 seeds × 500k per cell, ≈40 cells). Direction A grafts FB structure onto the scalar-TD grounding: the bilinear head (2); a z-conditioned forward map $\phi(s, a, \text{sg}\psi(g))$; a two-tower ϕ; sphere normalization; FB's orthonormality penalty; and the full stack. Direction B grafts scalar-TD ingredients into the measure framework (measure-only, task-normed residuals, rank-1 test projections, z-mixture sampling). A separate *annealing* family warm-starts from the converged 0.904 checkpoint and schedules loss mixtures (fully described in the companion document). *Findings.* Structure survives the grounding swap: bilinear **0.904**, z-cond ϕ 0.844, two-tower 0.864, full FB stack 0.871 (only ortho λ=1000 kills: 0.072). The identical architectures under the measure objective: **0.08–0.34**. Decisive: warm-start *at* the 0.93 solution and fade the scalar TD out ⇒ 0.05 — **the measure objective's fixed point on cube is bad; the objective is the whole gap.** Also: fitting the measure well is insufficient — the z-conditioned graft cut the measure TD residual 24× (6347→266) and moved success only to 0.255.

Q: Why is FB's task-inference interface also broken?

Experiment. E1a: take the trained bilinear-GCIQL critic with a z-conditioned actor ($c = \text{sg}[\psi(g)]$, group G4, d=512, 1M steps, 12 seeds at the anchor), and evaluate FB's zero-shot rule: relabel 50k dataset transitions with the task reward, infer $z = \sum_i r_i \psi(s'_i)$, rescale to the norm scale the actor saw in training, roll out with z only (the policy never sees g). Ablate the inference: moment vs whitened (GLS) vs exact $\psi(g)$; $\sqrt{d}$ vs matched norm. *Findings.* Moment inference **0.955** = the exact-z anchor. Norm hygiene is worth 0.29–0.46 (the $\sqrt{d}$ convention collapses two tasks); whitening is unnecessary and ill-conditioned (ψ covariance condition number $7{\times}10^{14}$). **The FB API is intact; only the grounding was broken.** (An earlier "inferred-z → 0.07" result was an on-policy sampling artifact: a badly conditioned policy rarely reaches the goal, so its own rollouts give a degenerate z.)

Q: Why is the bilinear win an out-of-distribution generalization story?

Experiment. The *OOD grid*: byte-identical start states and goals across all cells (serialized to disk), crossing start distribution (dataset states vs `env.reset()`, which samples arm poses the play policy never produces — median whitened NN-distance to the data 2.4× a held-out state's) with goal distribution (hindsight vs evaluation task): C1 (ID/ID), C2 (ID/OOD-goal), C4 (OOD-start/ID), C3 (both = the real benchmark). H=200, the environment's own success rule; the harness reproduces published numbers on C3 (MLP 0.747 vs published 0.717; bilinear 0.947 vs 0.904; FB 0.40 vs 0.42). 3 seeds. *Findings.* Fig. 1. (a) **FB fails in-distribution**: C1 = 0.43 while grasping 100% of episodes — post-grasp steering, not generalization. (b) The bilinear–MLP separation is the *goal* axis: the goal shift is free for bilinear ($\Delta{\approx}{+}0.06$) and costs the MLP 16–24 points; the start shift hurts both (−7 to −28) and separates nobody. (c) The MLP is the *better* critic fully in-distribution (C1: 0.98 vs 0.87). Goal generalization is real but cannot explain (c) — keep asking.

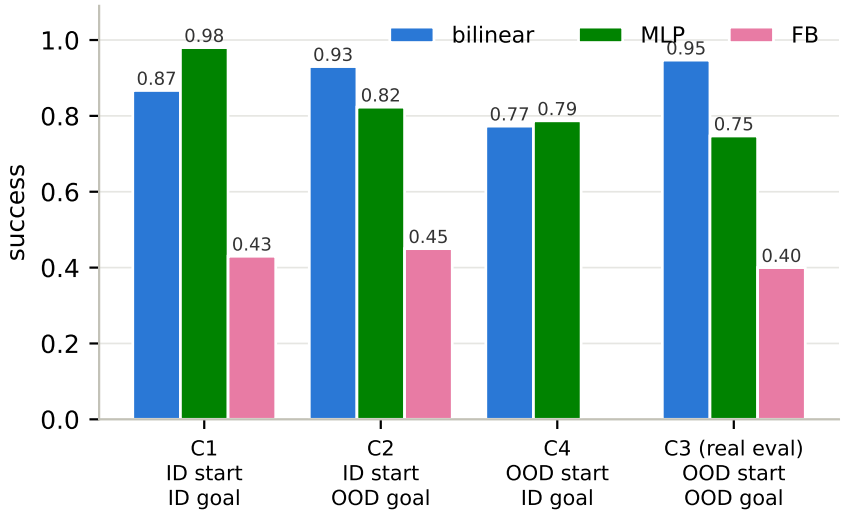


Figure 1: The OOD grid. FB's C1 uses the $z{=}B(g)$ route; its C4 was not run.

Q: Why is the bilinear critic simply a more accurate value function?
Experiment. The *counterfactual value probe*: reset the simulator to 200 in-hand grasp states (bit-exact prefix replay — snapshot/restore drifts because the IK-target integrator lives outside `mjData`), roll out the agent's own flow policy to success/failure, and measure the rank-AUC of the critic's value against the realized outcome. Actor held fixed across critics. A *decomposition probe* splits $Q_m = U_m(s,g) + A_m(s,a,g)$, $U_m = \mathbb{E}_{a\sim\mu} Q_m$ over 256 behavior-flow actions, and evaluates each part separately. *Findings.* MLP AUC **0.81**; bilinear **0.39**, below chance. The MLP's entire signal is in the action-independent baseline U (0.82); the bilinear's U is anti-calibrated (0.40); across states the two baselines are uncorrelated (Spearman -0.04) while the two *advantage* orderings partially agree ($+0.22$). **Not accuracy: the two heads allocate capacity to different parts of Q** — MLP to the baseline, bilinear to the action-dependent part. (This also revises an earlier program thesis: cross-state value fidelity is neither necessary nor what carries control.)

Q: Why what, then, does the bilinear critic do better?
Experiment. Reranking: freeze the trained critic; at every environment step draw N candidate actions from a frozen goal-agnostic BC flow μ and execute $\arg\max_i Q(s,a_i,g)$. C3 protocol, $N \in \{1..128\}$, 50 episodes/task. Controls: swap the candidate flow between agents; *fresh-actor* experiments train a new flow actor (or a new DDPG+BC actor) from scratch against the frozen critic (500k steps, optimizer masking verified bit-frozen). *Findings.* Fig. 2. Bilinear-as-ranker **0.91–0.96** at $N{=}32$ = its fully trained actor (0.947), *with no actor training*; declines to 0.85 at $N{=}128$ (more draws find the off-support peak, next rung). MLP-as-ranker **0.57–0.59** $<$ its trained actor (0.747). FB-as-ranker *declines with* N ($0.52 \to 0.17$): its Q actively misleads about actions. Fresh actors reproduce the gap (flow: 0.88 vs 0.68; co-adaptation refuted). **The bilinear critic's product is the on-support action ordering; the ordering is the policy.** Corroboration: action-dependence is $< 1\%$ of Q variance for every critic (bilinear 0.46%, FB 0.21%, MLP 0.10%) — the loss barely sees it; only the bilinear head turns it into usable structure.

Q: Why does the gradient-ascent actor lose with a critic whose ordering is near-perfect?
Experiment. Off-reference test: for the co-trained DDPG+BC agents, measure (i) the whitened nearest-neighbor distance from the deployed actor's action to a 256-action behavior-reference cloud,

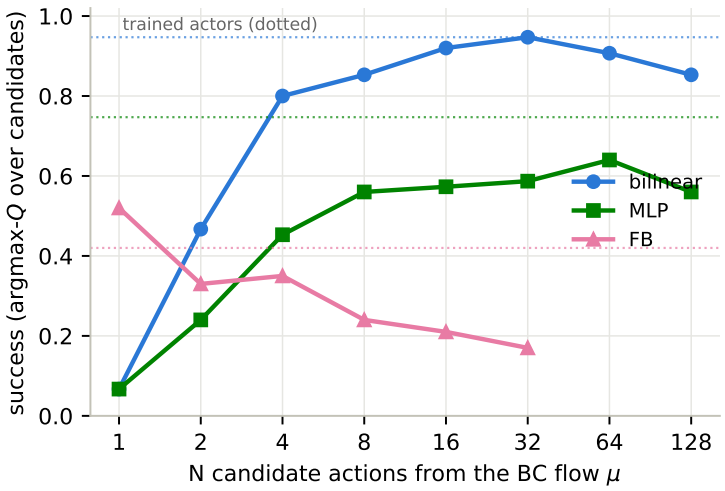


Figure 2: Frozen-critic reranking of N behavior-flow candidates (the mechanism figure).

(ii) $Q(a_{\text{DDPG}}) - \max_{a \in \text{ref}} Q(a)$ and the fraction of states where it is positive, (iii) action-gradient norms $\|\nabla_a Q\|$ (equal-slope vs peak discrimination). Fresh-DDPG+BC-vs-frozen-bilinear isolates intrinsicness. *Findings.* The bilinear critic values the fabricated action above *every* reference action 79% of the time (median +2.14); the MLP 6% (−0.26). Gradient norms are equal (3.18 vs 3.06): an off-support *maximum*, not a steeper landscape. The DDPG+BC actor departs the behavior manifold 10× further on bilinear (whitened distance 17.9 vs 1.76); a fresh DDPG+BC against the frozen critic lands at 0.52 (≈ co-trained 0.557 ≪ flow 0.88). **The critic's one flaw lives exactly where gradient extraction looks and anchored extraction never does** — producing the actor reversal (bilinear+flow 0.947 / bilinear+DDPG+BC 0.51 / MLP ≈0.73 either way).

Q: Why is the off-support peak intrinsic to the bilinear form?

Experiment. T3: recompute the peak statistic on the bilinear+DDPG+BC checkpoints of three domains (cube, antmaze-medium, scene; 512 dataset states × 5 eval goals; reference cloud = the dataset action at s plus the 64-NN states' actions in whitened observation space — no BC flow needed), plus a dataset-only action-coverage statistic (dimension-corrected against a same-size uniform null). *Findings.* Fig. 3. Cube 0.639 (replicating the μ-cloud version), antmaze **0.003**, scene 0.014 — the peak does not exist on antmaze, which is why bilinear+DDPG+BC *wins* there (0.837 vs MLP 0.695). The pre-registered *data-coverage* explanation is **refuted**: cube's conditional action manifold is only 1.11× narrower than antmaze's — far too small for a 0.639 → 0.003 flip. **The peak's location is a property of the domain's value geometry, not the data** (open thread, §3).

Q: Why does the bilinear head rank actions well while the MLP ranks barely above chance — is the MLP's action signal merely noisy?

Experiment. T1, *goal-smoothed reranking*: rerank with $\bar{Q}_K(s,a,g) = \frac{1}{K}\sum_j Q(s,a,g_j)$ where the g_j perturb the goal's cube-position dimensions by $\mathcal{N}(0,\sigma^2)$, $K \in \{1,8,32\}$, $\sigma \in \{2,4\}$ cm (= the success radius), 250 episodes/arm; control arm: g_j = random dataset goals (smoothing must be *local* to help). Gate: K=1 must reproduce the known rerank numbers (it does: 0.924 / 0.616). *Findings.* Fig. 4. Smoothing makes the MLP monotonically *worse* (0.616 → 0.408 at K=32, σ=4 cm); a 2–4 cm goal perturbation is already distribution shift for the MLP head (the same brittleness as its −16 to −24 goal-shift loss). Bilinear is invariant at 2 cm (0.912) and both collapse under the random-

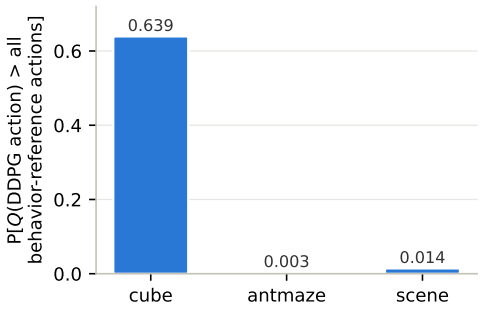


Figure 3: The off-support peak by domain.

goal control (0.148/0.040), validating the instrument. **The MLP's action signal is absent or systematically wrong, not noisy** — eval-time pooling cannot recover it. If cross-goal pooling is the bilinear mechanism, it acts at *training* time, through the shared ϕ basis. The causal control for that hypothesis is open (§3).

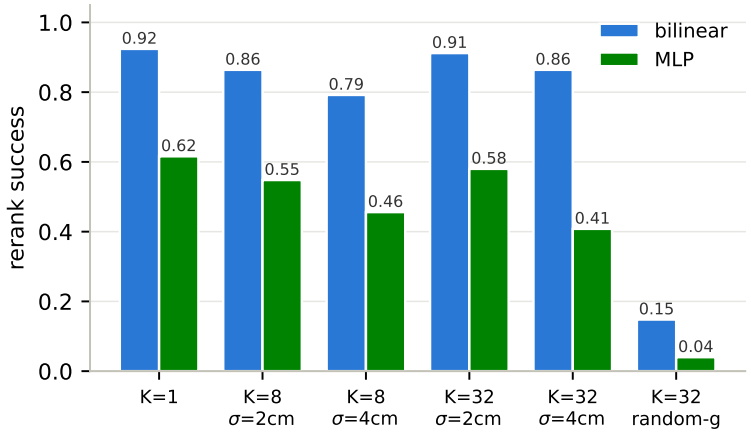


Figure 4: T1: eval-time goal smoothing does not rescue the MLP.

Q: Why is the factorization also responsible for training stability?

Experiment. Ablate the bottleneck: `bilinear_sg` replaces $\psi(g)$ with $\psi(s, g)$ (same parameter count up to the input layer; every other knob fixed; 3 seeds). T2 telemetry on the resulting checkpoints and on 50k-interval retrains of the stock heads: normalized feature co-adaptation $\mathbb{E}[\phi(s, a)^\top \phi(s', a')]$ (the DR3 statistic), per-leg norms, q extremes, on a fixed 4096-transition batch. *Findings.* 2/3 seeds diverge late (value loss $\times$10–30; Q crosses positive on a ≤ 0-bounded value); the stable seed lands at MLP level (0.69). The DR3 statistic does *not* discriminate (saturated $\approx$0.99 for all heads); the divergence lives in the **state-conditioned goal leg**: $\mathbb{E}\|\psi\|^2 = 1083/2355/3789$ across seeds, monotone in divergence severity, $q_{\max}$ to $+1.2\times10^3$, while ϕ stays sane. **Revised mechanism: global coupling as implicit regularization** — with $\psi(g)$, any score inflation must pass through globally shared features and is corrected by the rest of the data; $\psi(s, g)$ opens a local (s, g) pocket that expectile-TD's upward pressure (no output bound, no V target) inflates.

Q: Why how much rank does the win need, and which incidental knobs matter?

Experiment. Rank sweep $d \in \{5, 28, 50, 128, 512\}$ and self-ablations (single critic head; expectile

$\tau \in \{0.5, 0.9, 0.95\}$), 3–13 seeds each. *Findings.* d=5 collapses (0.12); $d \geq 28$ is flat (0.86–0.91): **rank is a floor, not a dial** — the task family's demands are low-rank; the win is not about tuning d. Single-head 0.874 (−3); τ=0.5 **0.920**, τ=0.95 0.874 — the expectile is not the ingredient either (τ=0.5 is plain SARSA-style expectile and works slightly better).

Q: Why does the effect transfer to other value learners and domains?
Experiment. The SOTA replication: swap MLP→bilinear inside GCIVL (AWR extraction), HIQL (hierarchical), and across antmaze/scene; 3–8 seeds per cell. *Findings.* Cube: GCIVL 0.675 vs 0.496 stock; HIQL improves only when the bilinear TD loss is routed *around* the shared subgoal representation (two-head split 0.35–0.42 vs shared 0.15–0.19; the published-tuning cell is confounded and was rerun at the correct `subgoal_steps`). Antmaze: bilinear+DDPG+BC **0.837** (vs 0.695); flow fails there for *both* critics (0.24–0.33) — an actor-side artifact (the goal-agnostic BC anchor fights direction commitment on a direction-multimodal marginal), currently being fixed by conditioning the anchor. Scene: bilinear loses with both actors (0.26–0.29 vs 0.52) and exhibits *no* off-support peak — a distinct, undiagnosed failure (§3). **The critic-side win transfers across value learners on cube; the extraction layer is what breaks portability.**

Q: Why summing up — why does bilinear GCIQL + flow work on cube?
Three verified clauses. (1) *Sound grounding*: scalar hindsight TD is contractive and task-weighted; the measure alternative is neither, and its fixed point here is bad. (2) *Capacity reallocation*: the factorization cannot ignore the $< 1\%$-variance action signal the way a monolithic net can — one shared ϕ serving every goal accumulates it into a near-perfect on-support ordering, paying with value calibration (irrelevant downstream), gaining free goal generalization and backup stability, and leaving one structured flaw (the off-support peak). (3) *Extraction match*: anchored extraction consumes exactly the ordering and never visits the flaw; gradient extraction does the opposite. Cube is where all three trades pay simultaneously (goal-shift-dominated benchmark; post-grasp ordering bottleneck; benign anchor).

3 The bedrock: open questions about bilinear Q-networks

These are the questions the chain could not answer; each carries its proposed experiment. This section is the live frontier of the document.

1. **Why does the low-rank constraint force a good action ordering?** The surviving hypothesis is training-time cross-goal pooling through the shared ϕ basis; the rivals (multimodality, direction-vs-distance content, eval-time denoising) are dead. *Queued (T5)*: (a) a *late-fusion* control — identical two towers, MLP readout on $[\phi; \psi]$ instead of the inner product (isolates the readout from the tower structure); (b) a parameter-matched MLP; (c) feature-rank (srank) trajectories of the MLP critic to test the rank-collapse reading of its near-chance ordering.

2. **Where does the off-support peak form?** Coverage is refuted; the working idea is quasimetric-residual relocation (a symmetric rank-d form cannot represent the asymmetric optimal goal-value; the residual must land somewhere, and its location follows the domain's value geometry). *Queued*: fit the peak's location against embedding-norm growth off-manifold ($\|\phi(s, a)\|$ along $a_{\text{ref}} \to a_{\text{DDPG}}$ rays), cube vs antmaze; a positive result yields a *predictive* account (and a test: norm-capped ϕ should shrink the peak).

3. **Why does the same head lose on scene?** No peak, both actors fail; hypotheses: combinatorial goal structure under a single rank-50 ψ; or the HIQL-style shared-rep poisoning. *Queued*: scene rerank N-sweep (in flight), then a Schaul-style split and a per-factor goal head.

4. **How low-rank is the *true* advantage?** $d \geq 28$ sufficing suggests the optimal action-ordering structure of cube is genuinely low-rank; is it? *Queued (T9)*: estimate $\text{rank}(A^*(s, \cdot, g))$ over the goal grid from counterfactual rollouts (atlas banks + simulator).

5. **Regime inversion vs the online literature.** Bilinear Value Networks (online, HER, DDPG) found the *state-conditioned* pairing $f(s, a)^\top \phi(s, g)$ essential; offline we find the opposite ($\psi(g)$ stable, $\psi(s, g)$ divergent). Is the difference the offline expectile backup (no bound + no V-target + local inflation pockets), or the data regime? *Queued*: $\psi(s, g)$ with an output bound / V-target — if it stabilizes at bilinear level, the divergence, not the pairing, is the offline story.

6. **Does the ordering quality have a clean information-theoretic signature?** The critic's U and A decouple (uncorrelated baselines, partially shared advantages). *Speculative*: measure mutual information between ϕ's top principal subspace and one-step controllable state deltas, bilinear vs MLP vs FB — a representation-level correlate of the reranking gap that could be computed at any checkpoint.

4 Related work: bilinearity and factored value representations in RL

(Living section; entries carry one-line relevance to our findings.)

Successor structure. Successor representations (Dayan 1993) and successor features $\psi^\pi(s, a)$ with $Q = \psi^\top w$ (Barreto et al. 2017, arXiv 1606.05312) are the original "value = features × task-vector" factorizations; USFA (Borsa et al. 2019, 1812.07626) adds policy-conditioning and GPI. FB (Touati & Ollivier 2021; Touati, Rapin, Ollivier 2023, 2209.14935) factorizes the successor *measure* $M \approx F^\top B\rho$ — our failed grounding. Our bilinear GCIQL is, in this taxonomy, successor features whose reward basis $\psi(g)$ is co-learned by scalar hindsight TD — "SFs without feature TD" (every explicit vector-TD estimator of the same structure collapsed in our lattice: 0.00–0.38).

Bilinear value heads. Bilinear Value Networks (Hong, Yang, Agrawal, ICLR 2022, 2204.13695) is the closest prior: $Q = f(s, a)^\top \phi(s, g)$ inside online DDPG+HER, with gains attributed to compositional goal generalization — matching our free-goal-shift finding, but with the pairing inverted relative to our stable configuration (§3, item 5). FLAP (2101.04750) shows shared-basis linear heads extrapolate across tasks in meta-RL. Low-rank structure of Q matrices: Yang et al. (ICLR 2020, 1909.12255) and Shah, Song, Yu (NeurIPS 2020) verify empirical control Q-matrices are near-low-rank and use rank restriction as a structured denoiser — independent support for "rank as a helpful prior, not a capacity loss" ($d \geq 28$ flat in our sweep).

Embedding-factored values from representation learning. Contrastive RL (Eysenbach et al. 2022, 2206.07568) learns $f(s, a, g) = \phi(s, a)^\top \psi(g)$ as a goal-conditioned critic by InfoNCE — structurally our head, trained by a different (contrastive, on-support-ranking-shaped) objective; C-learning (2011.08909) is its TD-flavored sibling. VIP (Ma et al., ICLR 2023, 2210.00030) derives an *implicit* time-contrastive objective from the Fenchel dual of goal-conditioned RL and produces a value as embedding distance $V(s; g) = -\|\varphi(s) - \varphi(g)\|$, pre-trained from passive video — evidence that value-shaped embedding geometry can be learned without TD at all; LIV (2306.00958) extends it to language–image goals. HILP (Park et al., ICML 2024, 2402.15567) embeds temporal distances isometrically ($\|\varphi(s) - \varphi(g)\| = d^*(s, g)$); METRA (Park, Rybkin, Levine, ICLR 2024, 2310.08887) maximizes $(\varphi(s') - \varphi(s))^\top z$ under a temporal-metric constraint — both are inner-product/distance

task-factored values with explicit geometric constraints where ours emerges from TD alone. Laplacian representations (Wu, Tucker, Nachum, ICLR 2019) and spectral decompositions of transition operators (Ren et al. 2022, 2208.09515) motivate the same low-rank structure from operator theory.

Quasimetric obstructions. $-V^*$ of goal-reaching is a quasimetric; symmetric inner products cannot represent it with low distortion — QRL (Wang, Torralba, Isola, Zhang, ICML 2023, 2304.01203), IQE (2211.15120), MRN (Liu, Zhu, Stone, AAAI 2023, 2208.08133). This is the principled source of our "irreducible, structured residual" — and the open question of *where* it lands (§3, item 2).

Theory: linear and low-rank MDPs. Linear MDPs (Jin, Yang, Wang, Jordan, COLT 2020, 1907.05388) make Q^π exactly linear in known features $\phi(s, a)$ for every policy; FLAMBE (Agarwal et al. 2020, 2006.10814) and representation learning for low-rank MDPs (Uehara, Zhang, Sun, ICLR 2022, 2110.04652) learn the factorization. Our setting is the "representation-learning" regime of this theory with $\psi(g)$ as a learned task embedding rather than a known reward parameter.

TD and representation dynamics. DR3 (Kumar et al., ICLR 2022, 2112.04716): TD implicitly grows feature dot-products of bootstrap-adjacent pairs (our T2 operationalization of it saturated and did not discriminate; the divergence we measure is a norm/pocket phenomenon instead). Implicit under-parameterization (Kumar et al., ICLR 2021, 2010.14498): bootstrapping collapses feature rank — the standing candidate for the MLP's absent action signal (srank probe queued). Self-predictive learning dynamics (Tang et al., ICML 2023, 2212.03319) analyze when two-tower TD-style objectives collapse — relevant to why our SF vector-TD cells die while scalar TD through the same structure lives. Norm/scale-constrained critics: HL-Gauss/Stop Regressing (Farebrother et al., ICML 2024, 2403.03950) and SimbaV2 (2502.15280) — bounded/structured critic outputs beating raw MSE regression under TD; our $\sqrt{d}$-scaled inner product (and the T10 scale-anchor episode) are instances of the same principle.

Extraction. One-step RL (Brandfonbrener et al., NeurIPS 2021), IDQL (2304.10573), SfBC (2209.14548), FQL (Park, Li, Levine 2025, 2502.02538), and the one-step≡critic-regularization equivalence (Eysenbach et al. 2023, 2307.12968) together predict our extraction findings: selection/anchored consumption of a frozen well-ordering critic is near-optimal offline; gradient consumption amplifies critic artifacts (BCQ, 1812.02900).

Benchmark. OGBench (Park, Frans, Eysenbach, Levine, ICLR 2025, 2410.20092). Published FB-on-cube anchors: TD-JEPA (2510.00739) reports FB 62.0 (RGB) on cube-single and ≤3 on cube-double; no published work diagnoses *why* — this document's chain is, to our knowledge, the only mechanism-level account.

5 Experiment queue (living)

T5 late-fusion + param-matched controls (open item 1) · peak-geometry norm probe (item 2) · scene rerank + split heads (item 3) · T9 true-advantage rank (item 4) · bounded-$\psi(s, g)$ (item 5) · srank trajectories (item 1c) · multi-env rerank N-sweeps (antmaze/scene; partially in flight) · RG-FB d512 ladder run · T10 500k reads.